

11_homework_Prall.rmd

First, set up the libraries and read in the data. We also will plot the explained variance by PC#

```
#install.packages("pcadapt")
#install.packages("patchwork")
#install.packages("viridis")
#install.packages("adegenet")
#install.packages("igraph")
library(pcadapt)
library(vcfR)
```

```
##
##      *****      ***   vcfR   ***      *****
##      This is vcfR 1.15.0
##      browseVignettes('vcfR') # Documentation
##      citation('vcfR') # Citation
##      *****      *****      *****      *****
```

```
library(adegenet)
```

```
## Loading required package: ade4

##
##      /// adegenet 2.1.10 is loaded //////////////////////////////////
##
##      > overview: '?adegenet'
##      > tutorials/doc/questions: 'adegenetWeb()'
##      > bug reports/feature requests: adegenetIssues()
```

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
library(readr)
library(patchwork)
library(viridis)
```

```
## Loading required package: viridisLite
```

```
meta <- read.csv("data/metadata.csv")
x <- read.pcadapt("data/salvelinus.vcf", type = "vcf")
```

```
## Warning in file2other(input, type, match.arg(type.out), match.arg(allele.sep)):
```

```
## Converter vcf to pcadapt is deprecated. Please use PLINK for conversion to bed
## (and QC).
```

```
## No variant got discarded.
```

```
## Summary:
```

```
##
```

```
## - input file:          data/salvelinus.vcf
```

```
## - output file:         /tmp/RtmpQU8rEB/fileea92034fab5c0.pcadapt
```

```
##
```

```
## - number of individuals detected: 267
```

```
## - number of loci detected: 24337
```

```
##
```

```
## 24337 lines detected.
```

```
## 267 columns detected.
```

```
pca <- pcadapt(x, K = 20)
plot(pca, option = "screeplot")
```

```
## Warning: The '<scale>' argument of 'guides()' cannot be 'FALSE'. Use "none" instead as
## of ggplot2 3.3.4.
```

```
## i The deprecated feature was likely used in the pcadapt package.
```

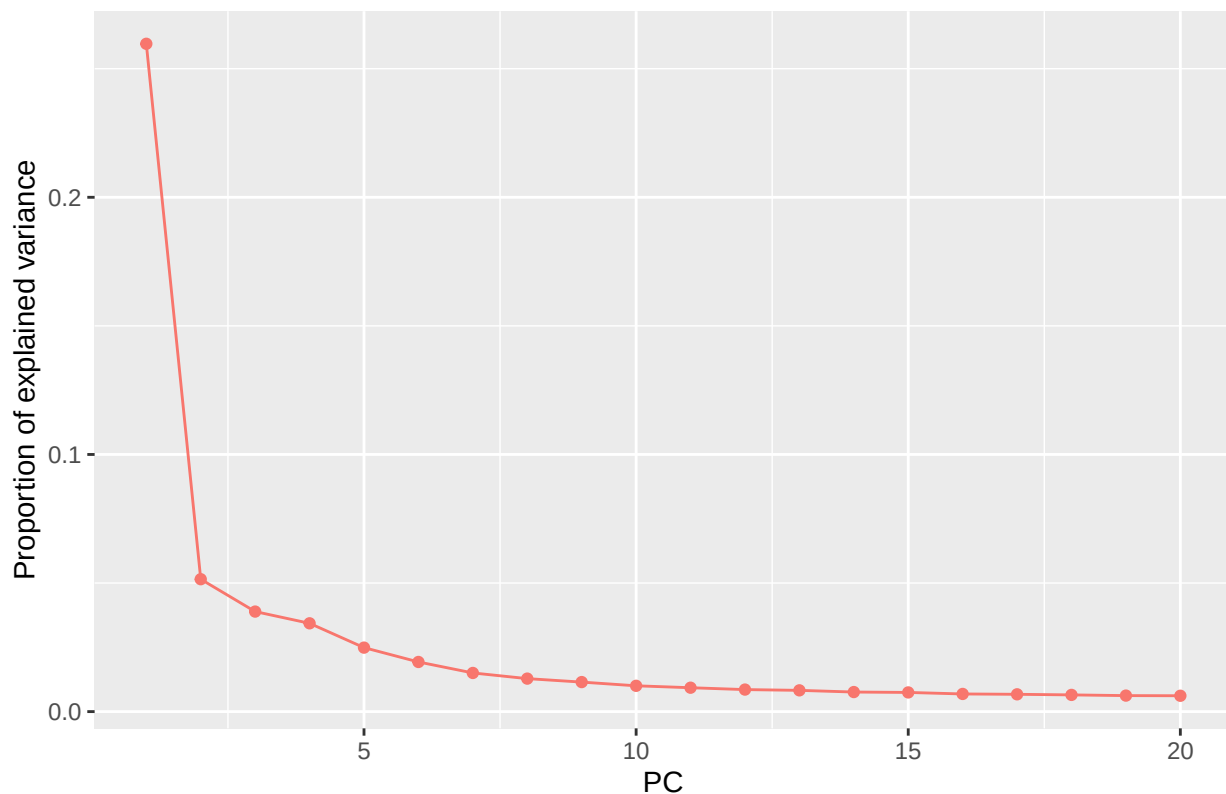
```
## Please report the issue at <https://github.com/bcm-uga/pcadapt/issues>.
```

```
## This warning is displayed once every 8 hours.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```

Scree Plot – K = 20



There is no obvious candidate, so we follow Meek et al, 2025 with K=4 These lines set that k, and identify outliers with p.adjust()

```
pca_4 <- pcadapt(x, K = 4)

alpha <- 0.05
padj <- p.adjust(pca_4$pvalues, method = "bonferroni")
outlier_idx <- which(padj < alpha)
neutral_idx <- which(padj >= alpha)
cat("Outlier loci:", length(outlier_idx), "\n")
```

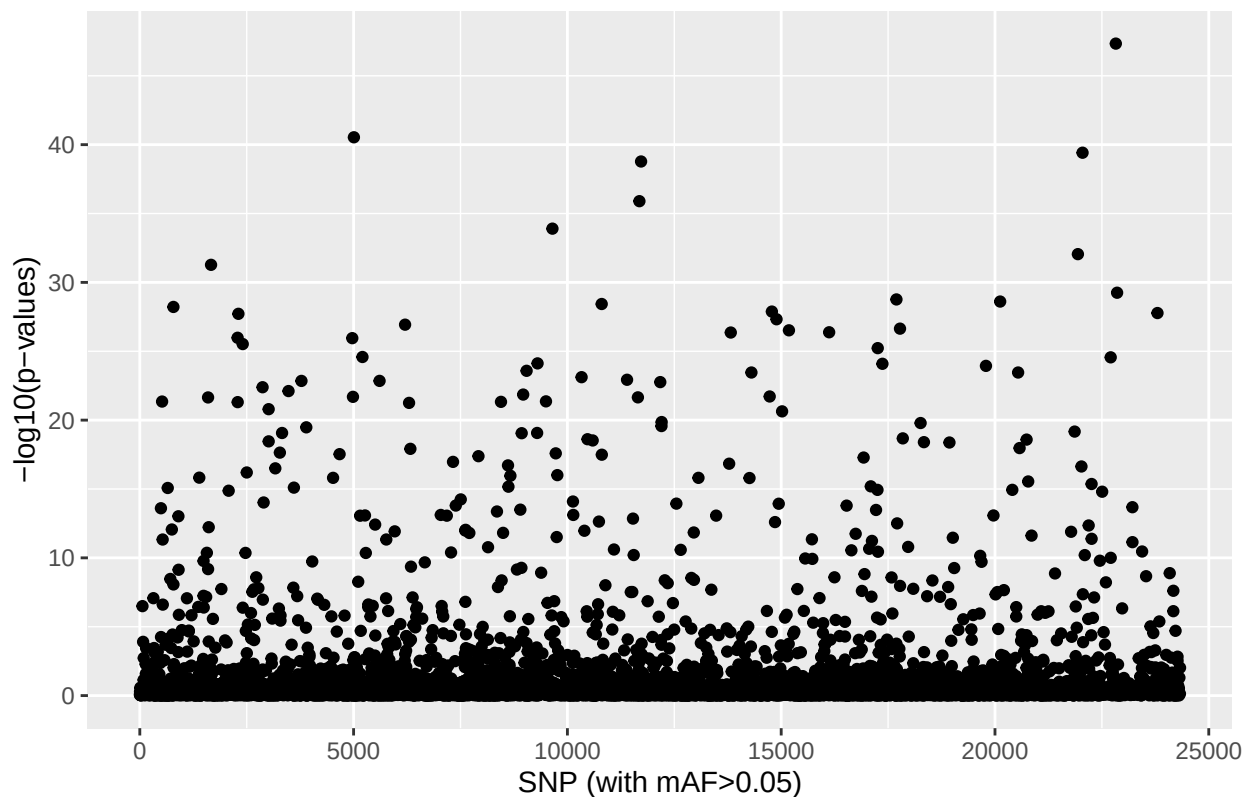
```
## Outlier loci: 306
```

```
cat("Neutral loci:", length(neutral_idx), "\n")
```

```
## Neutral loci: 5223
```

```
plot(pca_4)
```

Manhattan Plot



Reset this all for adegenet's implementation of PCA and create genind objects.

```
# Read in the files
vcf <- read.vcfR("data/salvelinus.vcf", verbose = F)
vcf_neutral <- vcf[neutral_idx, ]
vcf_outlier <- vcf[outlier_idx, ]

# turn into genind objects
dna_neutral <- vcfr2DNABin(vcf_neutral, unphased_as_NA = F, consensus = T, extract.haps = F)
```

```
## After extracting indels, 5223 variants remain.
```

```
gi_neutral <- DNABin2genind(dna_neutral)
dna_outlier <- vcfr2DNABin(vcf_outlier, unphased_as_NA = F, consensus = T, extract.haps = F)
```

```
## After extracting indels, 306 variants remain.
```

```
gi_outlier <- DNABin2genind(dna_outlier)
```

```
# Scale the genotypes
```

```
gi_neutral_scaled <- scaleGen(gi_neutral, NA.method = "mean", scale = F)
```

```
pca_neutral <- prcomp(gi_neutral_scaled, center = F, scale = F)
```

```
gi_outlier_scaled <- scaleGen(gi_outlier, NA.method = "mean", scale = F)
```

```
pca_outlier <- prcomp(gi_outlier_scaled, center = F, scale = F)
```

Transform the PCA objects for plotting

```
# lists into dfs
```

```
neutral_df <- data.frame(pca_neutral$x[,1:2]) %>% mutate(sample = rownames(.))
```

```
outlier_df <- data.frame(pca_outlier$x[,1:2]) %>% mutate(sample = rownames(.))
```

```
# join metadata
```

```
neutral_df <- left_join(neutral_df, meta, by = c("sample" = "Sample.ID"))
```

```
outlier_df <- left_join(outlier_df, meta, by = c("sample" = "Sample.ID"))
```

```
# shared axis limits
```

```
xlims <- range(c(outlier_df$PC1, neutral_df$PC1), na.rm = T)
```

```
ylims <- range(c(outlier_df$PC2, neutral_df$PC2), na.rm = T)
```

```
# percent variance explained
```

```
neutral_var <- 100 * summary(pca_neutral)$importance[2,]
```

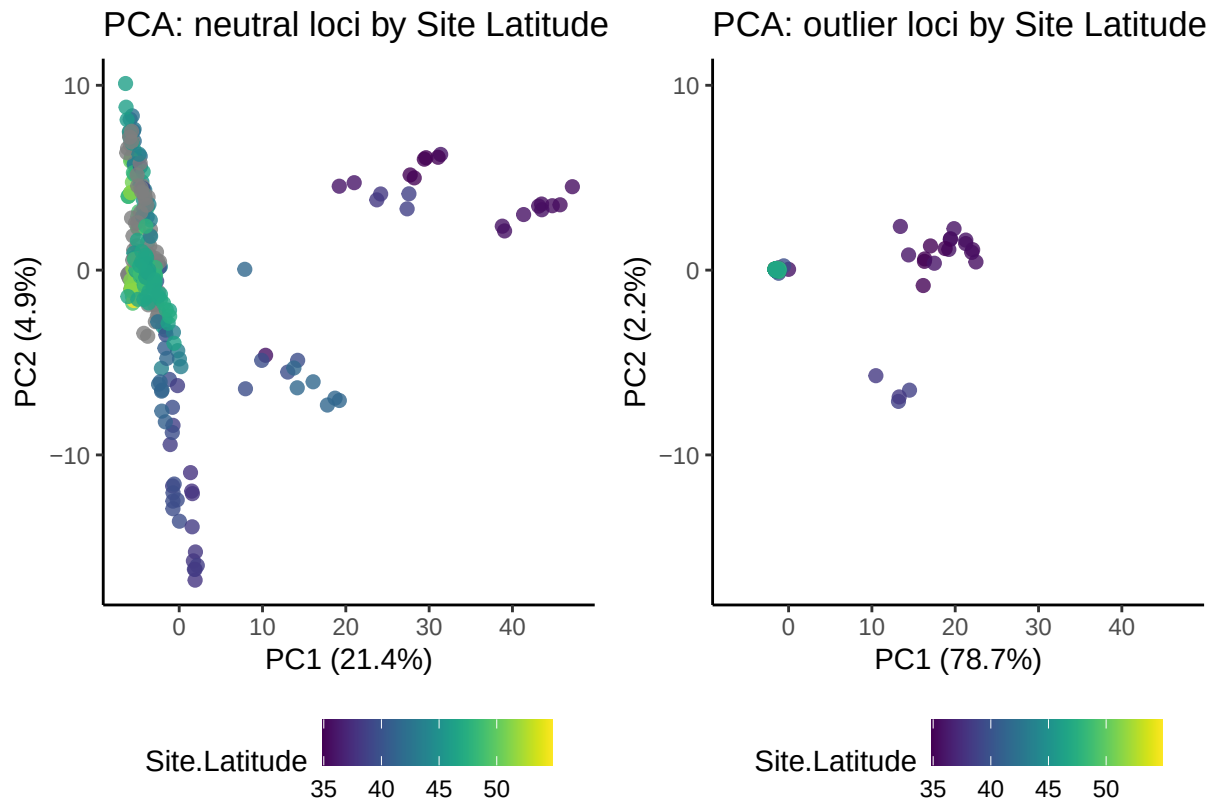
```
outlier_var <- 100 * summary(pca_outlier)$importance[2,]
```

```
# plot them both
```

```
p1 <- ggplot(neutral_df, aes(x = PC1, y = PC2, color=Site.Latitude)) +  
  geom_point(size = 2, alpha = 0.8) +  
  scale_color_viridis() +  
  coord_cartesian(xlim = xlims, ylim = ylims) +  
  theme_classic() +  
  theme(legend.position = "bottom") +  
  labs(title = "PCA: neutral loci by Site Latitude",  
       x = paste0("PC1 (", round(neutral_var[1], 1), "%)",  
       y = paste0("PC2 (", round(neutral_var[2], 1), "%)",  
       )
```

```
p2 <- ggplot(outlier_df, aes(x = PC1, y = PC2, color=Site.Latitude)) +  
  geom_point(size = 2, alpha = 0.8) +  
  scale_color_viridis() +  
  coord_cartesian(xlim = xlims, ylim = ylims) +  
  theme_classic() +  
  theme(legend.position = "bottom") +  
  labs(title = "PCA: outlier loci by Site Latitude",  
       x = paste0("PC1 (", round(outlier_var[1], 1), "%)",  
       y = paste0("PC2 (", round(outlier_var[2], 1), "%)",  
       )
```

```
p1 + p2
```



Here we see the two PCA plots for neutral and outlier loci. There are obviously more neutral loci than outliers. The neutral also appear to form roughly 4 clusters, while the outliers only form roughly 3. If we lower our resolution, there's one scattered group in the outliers, but the neutral loci still show a left, mostly vertical and "one" scattered cluster. In terms of latitude, we see the neutral loci from higher latitudes are focused only in the top left of the PCA chart, particularly at the top of the vertical cluster. For outliers, there aren't many high latitude points at all, but the one point that isn't visually navy blue also sits on the left of the chart, but more in the middle than those in the neutral PCA.

```
# lists into dfs
neutral_df <- data.frame(pca_neutral$x[,1:2]) %>% mutate(sample = rownames(.))
outlier_df <- data.frame(pca_outlier$x[,1:2]) %>% mutate(sample = rownames(.))

# join metadata
neutral_df <- left_join(neutral_df, meta, by = c("sample" = "Sample.ID"))
outlier_df <- left_join(outlier_df, meta, by = c("sample" = "Sample.ID"))

# shared axis limits
xlims <- range(c(outlier_df$PC1, neutral_df$PC1), na.rm = T)
ylims <- range(c(outlier_df$PC2, neutral_df$PC2), na.rm = T)

# percent variance explained
neutral_var <- 100 * summary(pca_neutral)$importance[2,]
outlier_var <- 100 * summary(pca_outlier)$importance[2,]

# plot them both
p1 <- ggplot(neutral_df, aes(x = PC1, y = PC2, color=Site.Longitude)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis() +
  coord_cartesian(xlim = xlims, ylim = ylims) +
```

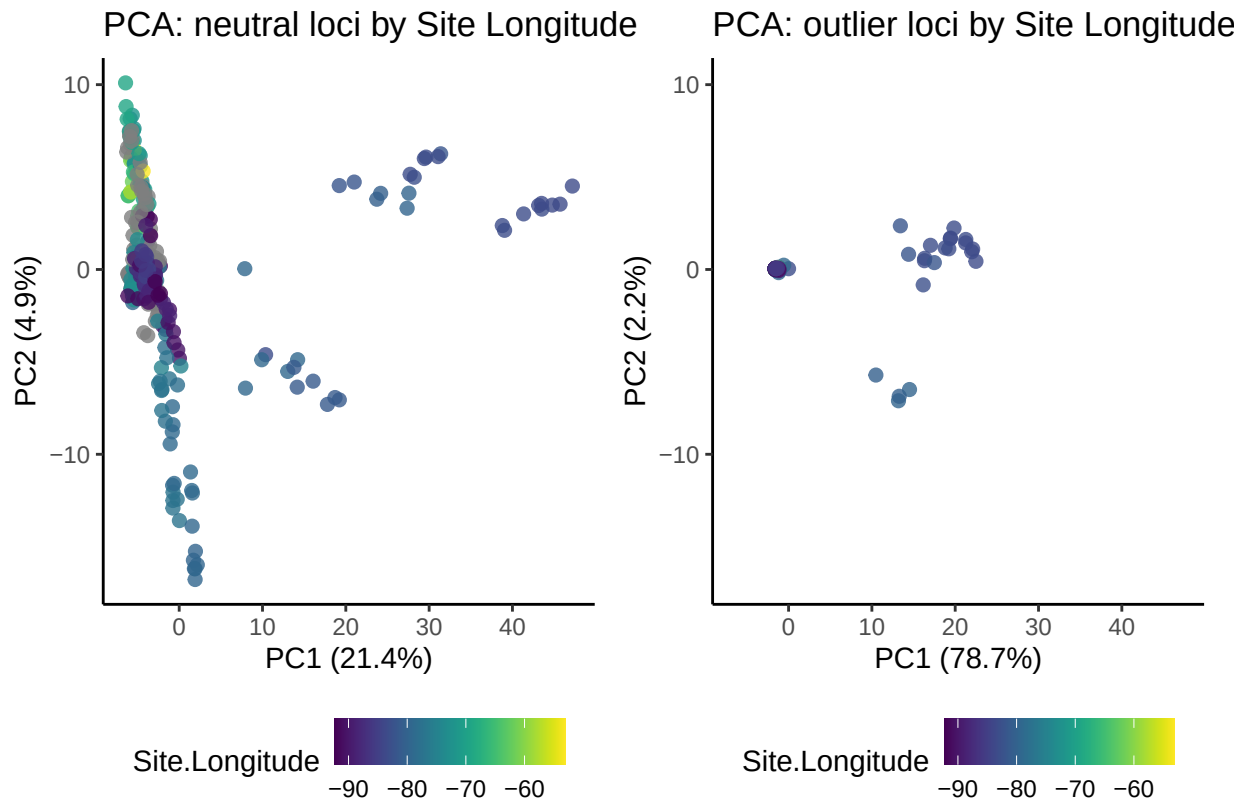
```

theme_classic() +
theme(legend.position = "bottom") +
labs( title = "PCA: neutral loci by Site Longitude",
x = paste0("PC1 (", round(neutral_var[1], 1), "%)",
y = paste0("PC2 (", round(neutral_var[2], 1), "%)",
)

p2 <- ggplot(outlier_df, aes(x = PC1, y = PC2, color=Site.Longitude)) +
geom_point(size = 2, alpha = 0.8) +
scale_color_viridis() +
coord_cartesian(xlim = xlims, ylim = ylims) +
theme_classic() +
theme(legend.position = "bottom") +
labs( title = "PCA: outlier loci by Site Longitude",
x = paste0("PC1 (", round(outlier_var[1], 1), "%)",
y = paste0("PC2 (", round(outlier_var[2], 1), "%)",
)

p1 + p2

```



Compared to the latitude map, the longitude map does not show the same patterning. For outlier loci, all points seem to fall above 80, so in trying to distinguish the shades of blue, it appears as if there is no correlation here between PCs and longitude. For neutral loci, the yellows and lighter greens, which correspond to lower longitudes, are mostly still in the upper left, but the blue-greens (medium longitudes) are distributed fairly evenly across the PCA plot. Additionally, there is a dense cluster of dark blue in the middle of the left cluster, blurring any correlation. However, the lack of patterning compared to the latitude plot makes sense, as latitude has a stronger correlation to climate traits (and thus selective forces) which longitude generally does not, particularly on a global scale.